REPORT 5 - NOSE CONE 720 MM HEATING, DRAG & LOADS

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Weisshorn – 720 mm Nose Cone: Thermal Profile, Drag A vs B and Loads

EPFL ROCKET TEAM

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1 Summary

What this report is for. It gives the thermal profile, the drag and the air loads of the two candidate 720 mm nose cones. It compares design A with design B, and gives the structure team (ST) what it needs to size the shell, the aluminium tip insert and the joint to the body tube. ST computes the stresses from these loads.

The two noses. Both are 720 mm long and 180 mm in diameter, with a 4.5 mm rounded tip. Their geometry is in `2027_C_FD_NOSE-CONE-720MM_GEOMETRY`.

	Design A (baseline)	Design B (backup)
Shape	LV-Haack	Von Kármán
3U payload	inside the nose	elsewhere in the rocket
Curved part + cylindrical coupler	679.0 + 41 mm	700.0 + 20 mm

Main findings

1. **Thermal profile (Section 3).** The shell is hottest just behind the tip and cools along the nose. Its **outer face** stays below the 120 °C of the DP490 bond: at most 88 °C on a nominal day and 117 °C on a hot day, so the hot-day margin is small. Its **inner face** is above the 67 °C of Req_10 over the whole length, already on a nominal day (69 to 81 °C); so is the air inside.
2. **A against B, heating.** Both designs heat up the same way: the shell temperatures differ by less than 1 °C. Only the tip insert of B is 5–6 °C cooler.
3. **A against B, drag (Section 4).** Below Mach 1 both noses have the same drag. Above Mach 1 design B has 13–15 % less: 237.6 N against 282.6 N at burnout (Fluent). B reaches 8982 m against 8887 m for A.
4. **The largest structural load comes from the nose being sealed.** The air inside stays at the pad pressure (101.3 kPa), while the outside pressure falls to 31.3 kPa at apogee. The shell is then pushed outward by about 70 kPa, and the nose is pulled off the joint with up to 1802 N, close to the 2000 N of Req_07. Vent holes would remove this load.
5. **The highest inward pressure is on the tip at burnout**: up to 177 kPa on the rounded tip only. Further back it is at most 43 kPa (A) and 22 kPa (B), and about 10 kPa or less beyond 50 mm.
6. **Side force**: 118 N per degree of angle of attack at burnout (hot day), acting 295 mm (A) or 347 mm (B) behind the tip.

Table 1: Values to design for. Each value is the worst case over the flight. The hot day is the worst case for every quantity. These are flight values, without a factor of safety, and air loads only: the inertial loads and the recovery load (Req_07) are not included. ✓: below the limit; ✗: above it.

	Unit	Design A	Design B	When	Limit
Temperatures (Section 3)					
Shell outer face, 10 mm (hottest station)	°C	116	117	9.4 s	120 (DP490) ✓
Shell outer face, shoulder	°C	98	98	10.2 s	120 (DP490) ✓
Shell inner face, 10 mm (hottest station)	°C	108	109	11.2 s	67 (Req_10) ✗
Shell inner face, shoulder	°C	94	95	12.2 s	67 (Req_10) ✗
Air inside the nose (upper bound)	°C	98	99	12 s	67 (Req_10) ✗
Aluminium tip insert	°C	85	79	12–13 s	
Drag and forces on the joint (Sections 4 and 5)					
Drag (compression)	N	310	260	burnout	
Pull-off force of the sealed nose (tension)	N	1788	1802	apogee	Req_07: 2000
Side force per degree of angle of attack	N/°	117	118	burnout	
Where the side force acts, from the tip	mm	295	347	always	
Pressure on the shell (Section 5)					
Inward pressure on the rounded tip	kPa	175	177	burnout	
Inward pressure behind the tip ($x > 2$ mm)	kPa	43	22	burnout	
Outward pressure on the shell	kPa	70	73	coast to apogee	

Where the numbers come from. Eight Fluent CFD runs of the two noses (2D, no angle of attack) at three moments of the climb, and the team's Python flight model, calibrated on these runs, for the rest of the flight and for the hot day. Appendix A explains the method. The data behind every figure are in the CSV files of `report_720/data/`.

2 Flight cases, load cases and conventions

This section defines the cases and the signs used in the rest of the report.

2.1 Two flight cases

Both designs are flown with the team's Python flight model (report 2), in two cases:

- **Nominal day**: standard atmosphere, 15 °C on the pad, propellant mass from the total impulse.
- **Hot day**: 30 °C on the pad and the LVO masses. The rocket is lighter and faster, so the air loads and the heating are higher. The hot day is the worst case for every value in this report.

Speed and dynamic pressure peak at burnout, 9.1 s after lift-off (Figure 1).

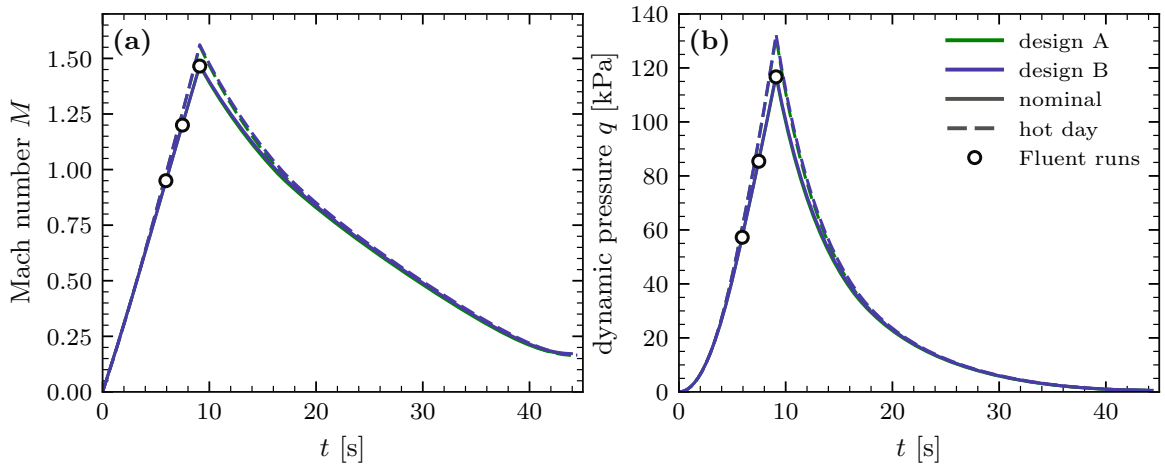


Figure 1: Mach number (a) and dynamic pressure (b) during the flight. Solid: nominal day; dashed: hot day. The circles mark the three Fluent runs.

2.2 Three load cases

Three moments of the flight cover the largest air loads (Table 2).

Table 2: The three load cases. Values for design A on a nominal day; design B differs by less than 0.5 s. On a hot day burnout is reached at $M = 1.56$ and $q = 131.4$ kPa.

Load case	Time [s]	Mach	Dynamic pressure [kPa]	Outside pressure [kPa]	Critical for
LC1, burnout	9.1	1.47	116.8	77.7	inward pressure on the tip, drag, side force and heating
LC2, transonic	5.9	0.95	57.6	90.7	the flow pattern changes near Mach 1; loads lower than at LC1
LC3, apogee	43.7	0.17	0.6	31.3	lowest outside pressure: outward pressure and pull-off force of the sealed nose

2.3 The sealed nose

The nose has no vent holes, so the air inside keeps the pressure it had on the pad, $p_{pad} = 101.3$ kPa. When the shell heats this air, its pressure rises:

$$p_{in} = p_{pad}\,\frac{T_{in}}{T_{pad}}. \tag{1}$$

The temperature of the inside air T_{in} is not computed. Two bounds are used instead:

- **cold inside air**, $T_{in} = T_{pad}$: lowest inside pressure, so the largest *inward* load;
- **heated inside air**, T_{in} = mean temperature of the inner face of the shell: highest inside pressure, so the largest *outward* load.

Each table says which bound it uses.

2.4 Sign conventions

Figure 2 shows the conventions used everywhere in this report.

- x is the distance from the tip along the axis. Four stations are followed during the flight: the tip, 50 mm, mid-length (half the curved part) and the shoulder (end of the curved part, where the coupler starts).
- The **net pressure** on the shell is $\Delta p = p_w - p_{in}$: the pressure on the outside of the wall minus the pressure inside. $\Delta p > 0$ pushes the shell **inward**; $\Delta p < 0$ pushes it **outward**.
- An **axial force** > 0 is compression: it pushes the nose aft onto the joint. An axial force < 0 is tension: it pulls the nose forward, off the joint.

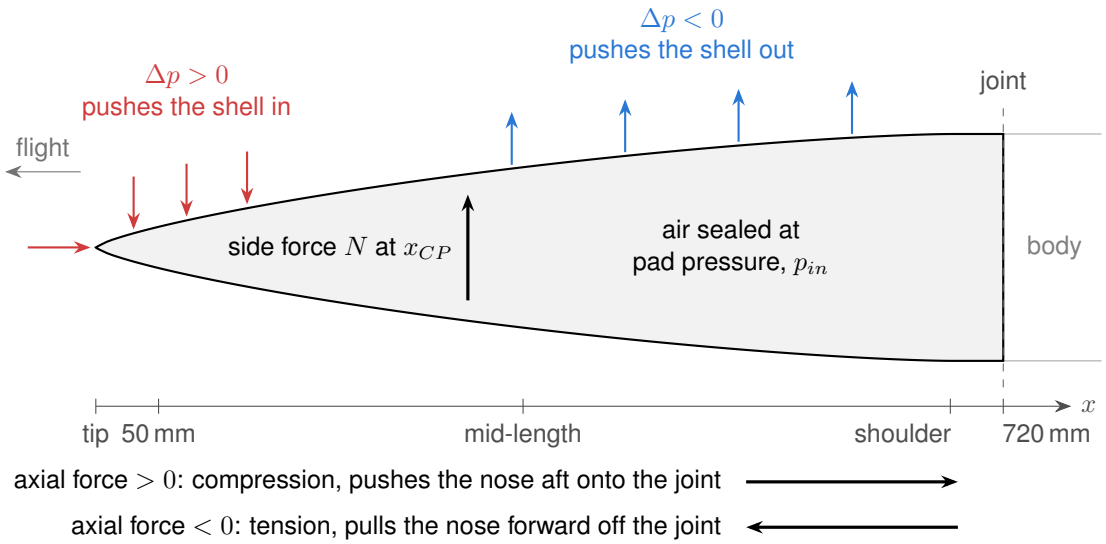


Figure 2: Sketch of the nose with the conventions used in all tables and figures.

3 Thermal profile

In short

- At burnout the air stopped at the tip reaches the stagnation temperature, 118–119 °C on a nominal day and 154–156 °C on a hot day. Along the rest of the nose the air drives the wall towards 380–383 K (about 107–110 °C).
- The shell is hottest just behind the tip and cools along the nose (Figure 9). The **outer face** stays below the 120 °C of the DP490 bond everywhere: at most 88 °C on a nominal day, but 117 °C on a hot day, just behind the tip, a margin of only a few degrees.
- The **inner face** goes above the 67 °C of Req_10 **over the whole length**, already on a nominal day: 69 to 81 °C (nominal) and 94 to 109 °C (hot day). So does the air inside.
- **A and B heat up the same way**: the shell temperatures differ by at most 0.8 °C. Only the tip insert of B is 5–6 °C cooler.

3.1 What Fluent shows

Burnout is the hottest moment of the flight: the rocket is at its highest speed, $M = 1.465$. Each design was run twice at this point:

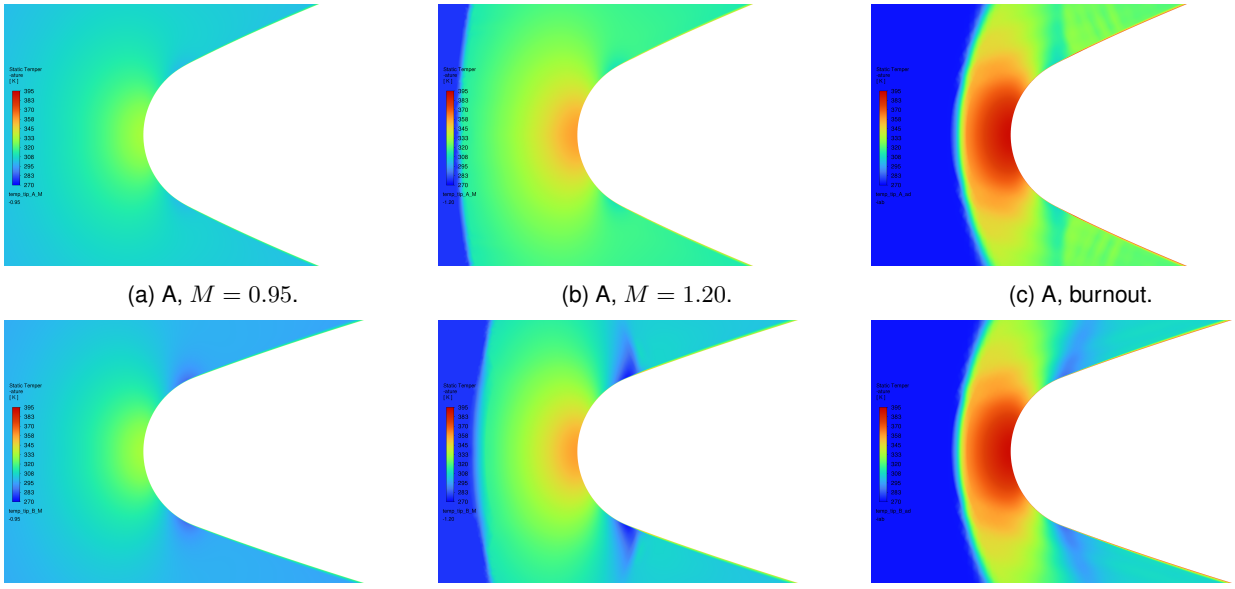
- with an **adiabatic wall** (no heat goes into the wall): the wall then takes the temperature the air drives it towards, the adiabatic wall temperature T_{aw};

- with the **wall held at 300 K**: this gives the heat flux q_w into a cold wall, and from the two runs the heat-transfer coefficient $h = q_w/(T_{aw} - 300\,\mathrm{K})$.

The air around the tip. The air that hits the rounded tip is stopped and heats up to the stagnation temperature. Figure 3 follows this through the climb:

- at $M = 0.95$ there is no shock yet: the air warms up gradually as it approaches the tip, to about 333 K;
- at $M = 1.20$ a bow shock stands ahead of the tip, and the air behind it reaches about 359 K;
- at burnout the shock sits closer to the tip and the air reaches about 391 K, the hottest moment of the flight.

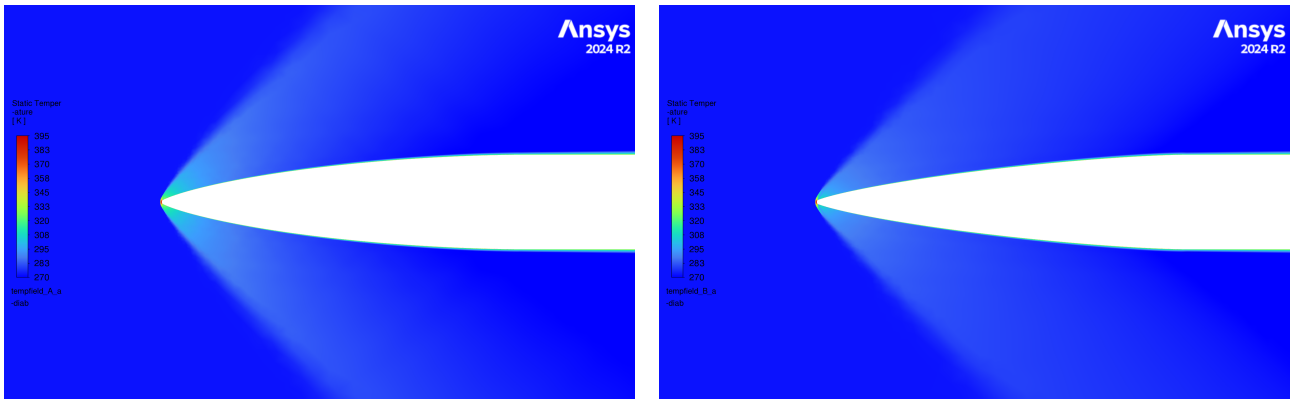
The two tips look the same at every point: the tip radius is the same, and so is the heating. Further back, the hot air stays in a thin layer along the wall, the boundary layer (Figure 4).



(a) A, $M = 0.95$. (b) A, $M = 1.20$. (c) A, burnout.

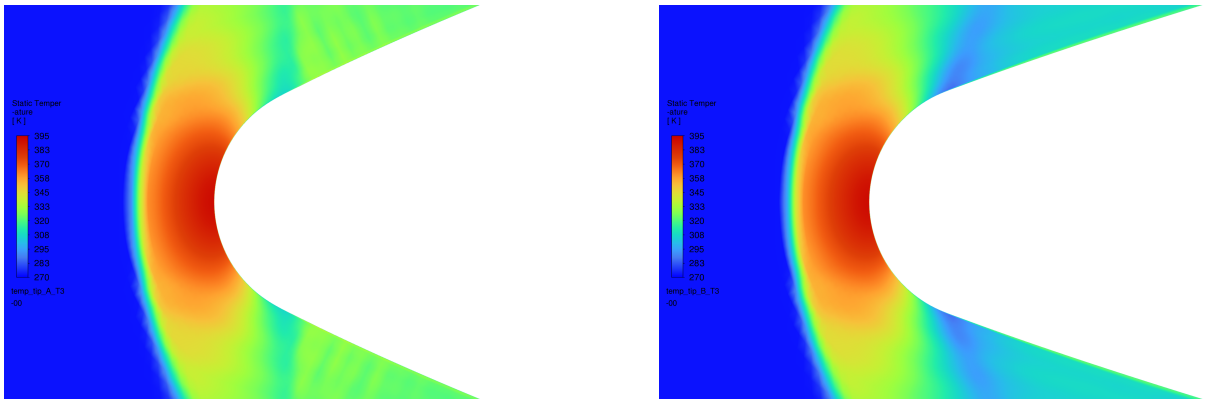
(d) B, $M = 0.95$. (e) B, $M = 1.20$. (f) B, burnout.

Figure 3: Air temperature around the tip at the three Fluent points of the climb (left to right: $M = 0.95$, 1.20 and burnout, $M = 1.465$), adiabatic wall; top: design A, bottom: design B. Same colour scale in all: 270 K (blue) to 395 K (red). The air at the tip heats up from about 333 K to 359 K and 391 K.



(a) Design A. (b) Design B.

Figure 4: Air temperature around the whole nose at burnout, adiabatic wall (Fluent). The hot air stays in a thin layer along the wall; the light band is the air heated by the bow shock. Colour scale: 270 K (blue) to 395 K (red).



(a) Design A. (b) Design B.

Figure 5: Air temperature around the tip at burnout with the wall held at 300 K (Fluent). The cold wall cools the air next to it: this temperature drop at the wall is the heat going into the nose. Colour scale: 270 K (blue) to 395 K (red).

Heating along the wall. Figures 6–8 give the heating along the wall of both designs, and Table 3 sums it up.

- The **wall temperature** T_{aw} is 391.3 K at the tip and 380–383 K further back: the whole nose is driven to about 107–110 °C at burnout.
- The **heat flux** and h are highest at the tip and fall along the nose: h goes from about 1900 W/(m^{2}K) at the tip to about 1000 W/(m^{2}K) at 50 mm and about 550 W/(m^{2}K) at the shoulder. So the front of the shell heats up fastest.
- A and B differ only over the first 10 cm, and by little. The heating does not separate the two designs.

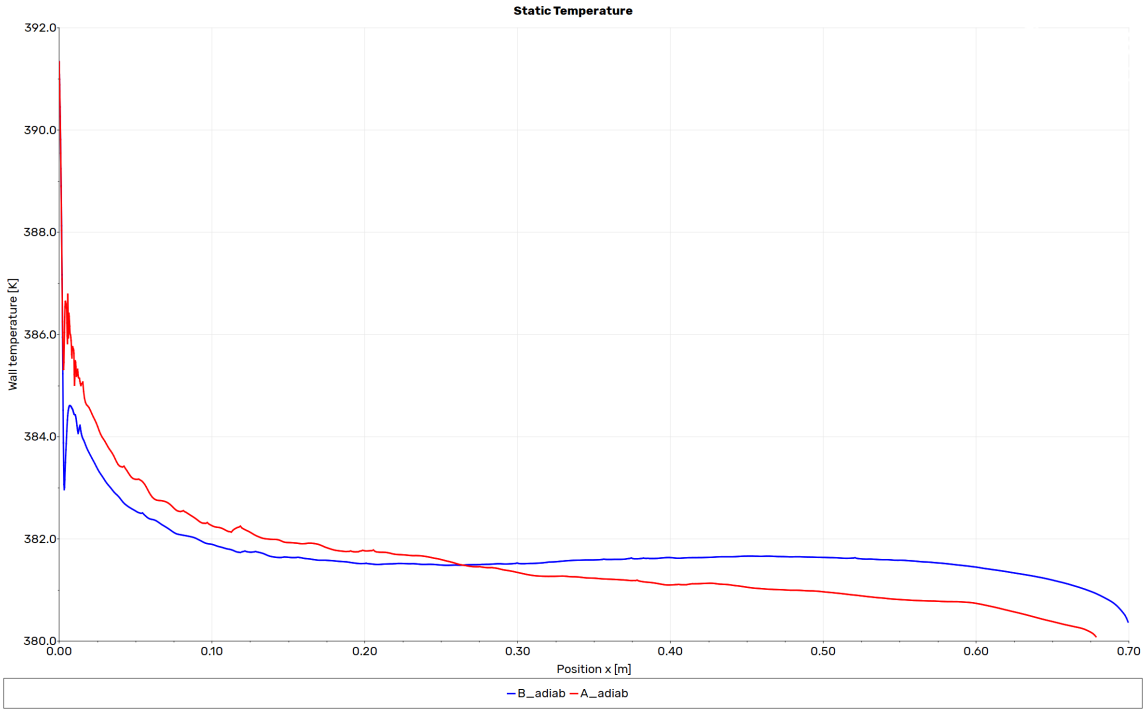


Figure 6: Adiabatic wall temperature along the nose at burnout, design A (red) and B (blue), Fluent. This is the temperature the air drives the wall towards. The ripple over the first 15 mm is a mesh effect, where the fine tip cells meet the wall cells.

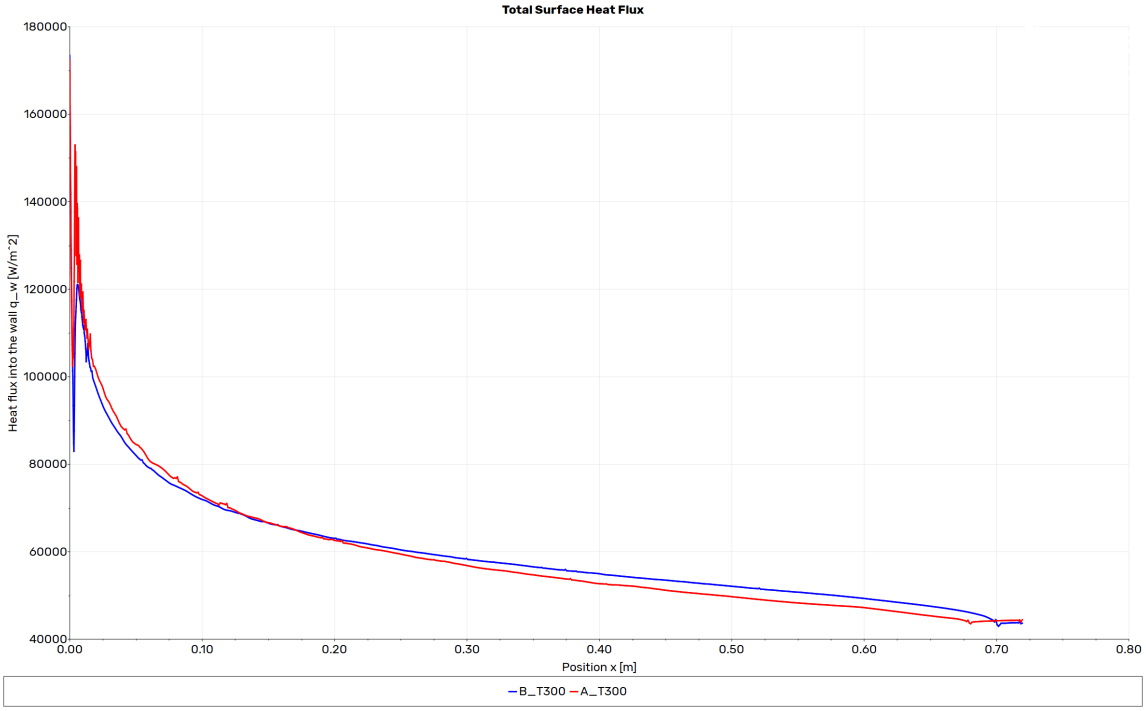


Figure 7: Heat flux into a wall held at 300 K along the nose at burnout, design A (red) and B (blue), Fluent.

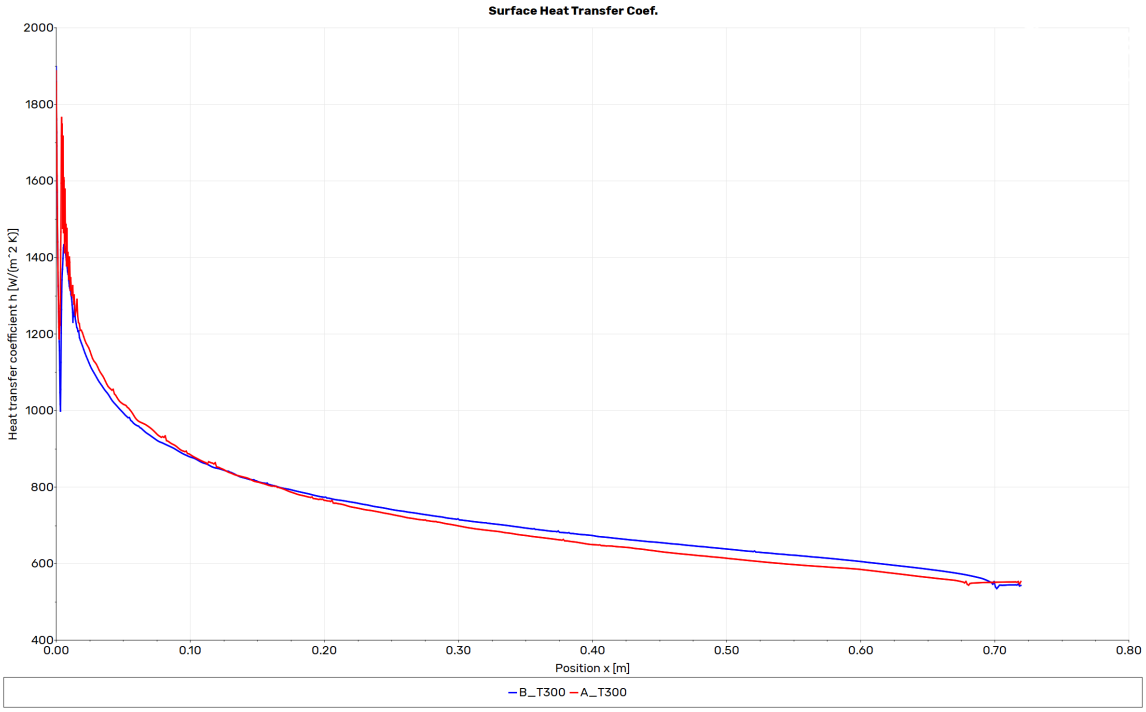


Figure 8: Heat-transfer coefficient $h = q_w/(T_{aw} - 300\,\mathrm{K})$ along the nose at burnout, design A (red) and B (blue), Fluent. This is the input of the shell model.

Table 3: Heating at burnout from Fluent, nominal day.

	Unit	Design A	Design B
Stagnation temperature of the air	K	391.6	391.6
Wall temperature (adiabatic) at the tip	K	391.3	391.3
Wall temperature (adiabatic) beyond 50 mm	K	380–383	380–383
Heat flux at the tip, wall at 300 K	kW/m^2	172.5	173.4
Heat-transfer coefficient h at the tip	W/(m^{2}K)	1888	1899
Heat-transfer coefficient h at the shoulder	W/(m^{2}K)	546	539

3.2 Temperature profile along the nose

Fluent gives the heating at burnout only. To follow the temperatures through the whole flight, the Python model of report 1 is used, with its heating calibrated on Fluent (Appendix A.4). It computes the heat conduction through the 1.2 mm CFRP shell at stations from 10 mm to the shoulder, heated on the outside. Two simplifications make the shell temperatures slightly high: no heat is lost on the inside, and over the first 80 mm the aluminium tip insert under the shell is ignored.

Figure 9 gives the **highest temperature reached during the flight** at every point of the nose, and Figure 10 shows **when and where** the shell is hot.

- Both faces are hottest near the tip and cool towards the shoulder, because h falls along the nose.
- The outer face stays below the DP490 limit everywhere. The inner face is above Req_10 everywhere.
- The inner face lags the outer face by about 2 s, because the heat needs time to cross the CFRP. Req_10 is exceeded from about 9 s to 14 s on a nominal day, and from 8 s to 19 s on a hot day.

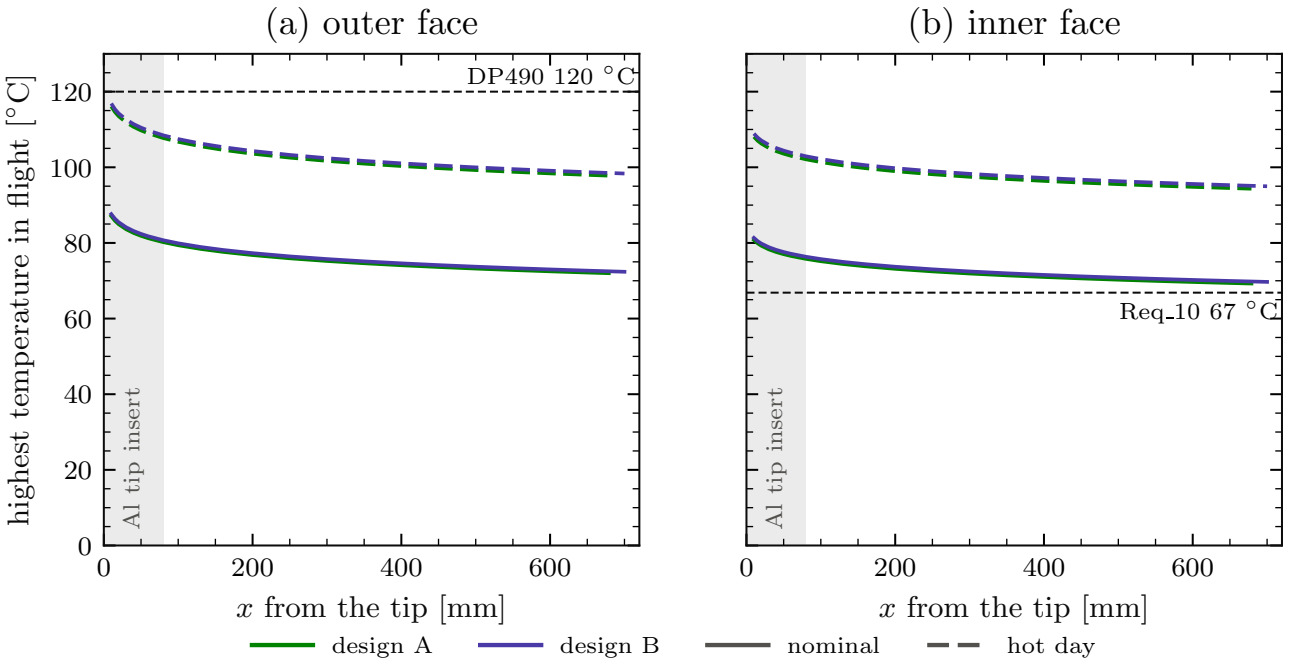


Figure 9: Highest temperature of the shell during the flight, along the nose: (a) outer face, (b) inner face. Design A in green, B in purple; solid: nominal day, dashed: hot day. Grey band: the first 80 mm, where the aluminium tip insert sits under the shell.

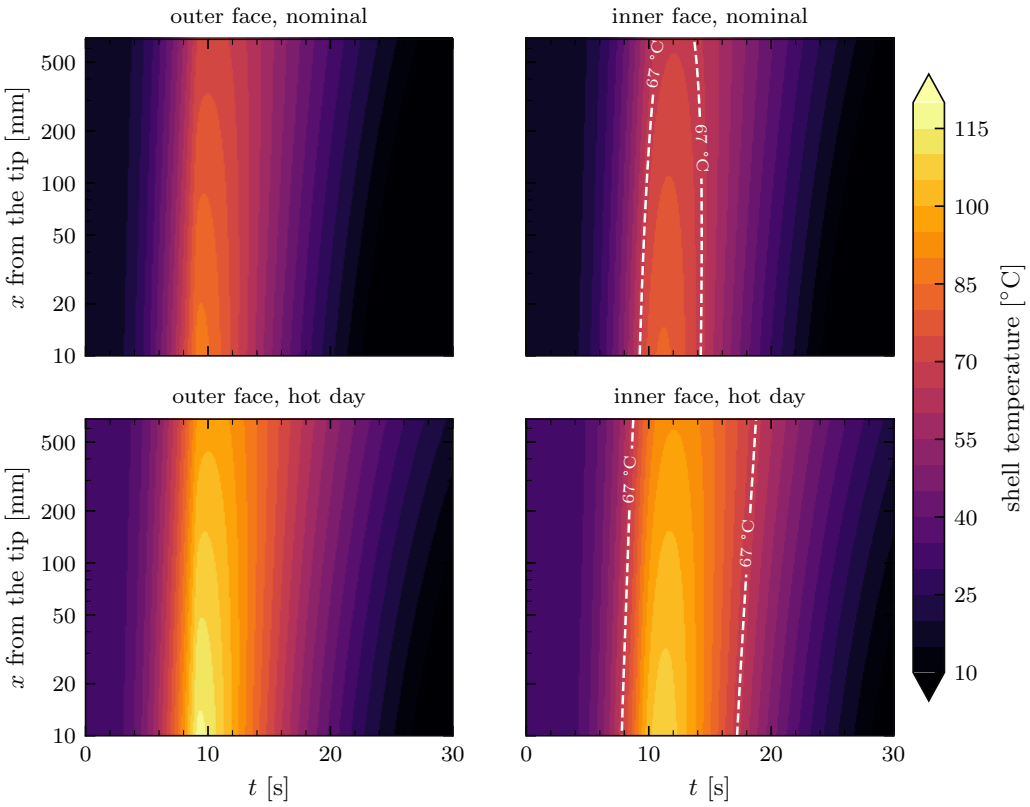


Figure 10: Shell temperature of design A against time and position. Left: outer face; right: inner face. Top: nominal day; bottom: hot day. White dashed line: 67 °C (Req_10) on the inner face. The outer face stays below 120 °C (DP490), so it has no line. Design B is the same within 0.8 °C.

Table 4: Highest shell temperature during the flight at four stations [°C]. ✓: below the limit; ✗: above it.

	Design A		Design B		
	Nominal	Hot day	Nominal	Hot day	Limit
Outer face					
20 mm	85	113	85	114	120 (DP490) ✓
50 mm	82	110	82	110	120 (DP490) ✓
Mid-length	75	101	75	102	120 (DP490) ✓
Shoulder	72	98	72	98	120 (DP490) ✓
Inner face					
20 mm	79	106	80	107	67 (Req_10) ✗
50 mm	77	104	78	104	67 (Req_10) ✗
Mid-length	71	97	72	98	67 (Req_10) ✗
Shoulder	69	94	70	95	67 (Req_10) ✗

3.3 Temperatures during the flight

Figure 11 follows the parts that matter most for ST through the flight: the aluminium tip insert, the shell at 50 mm (inside the region of the insert) and the air inside the nose. The peaks of the tip insert and the inside air are in Table 5, those of the shell in Table 4.

- The outer face of the shell peaks just after burnout (9.6 s), the inner face 2 s later.
- The tip insert, with its larger heat capacity, peaks later, at 12–13 s, and cools slowly.

- The tip insert of design B gets 20 % less heat than A's (its front is thinner, so there is less surface in the first 80 mm) and stays 5–6 °C cooler.
- The air inside follows the inner face of the shell (upper bound). Whether Req_10 applies to the air or to the inner wall is still open with SE.

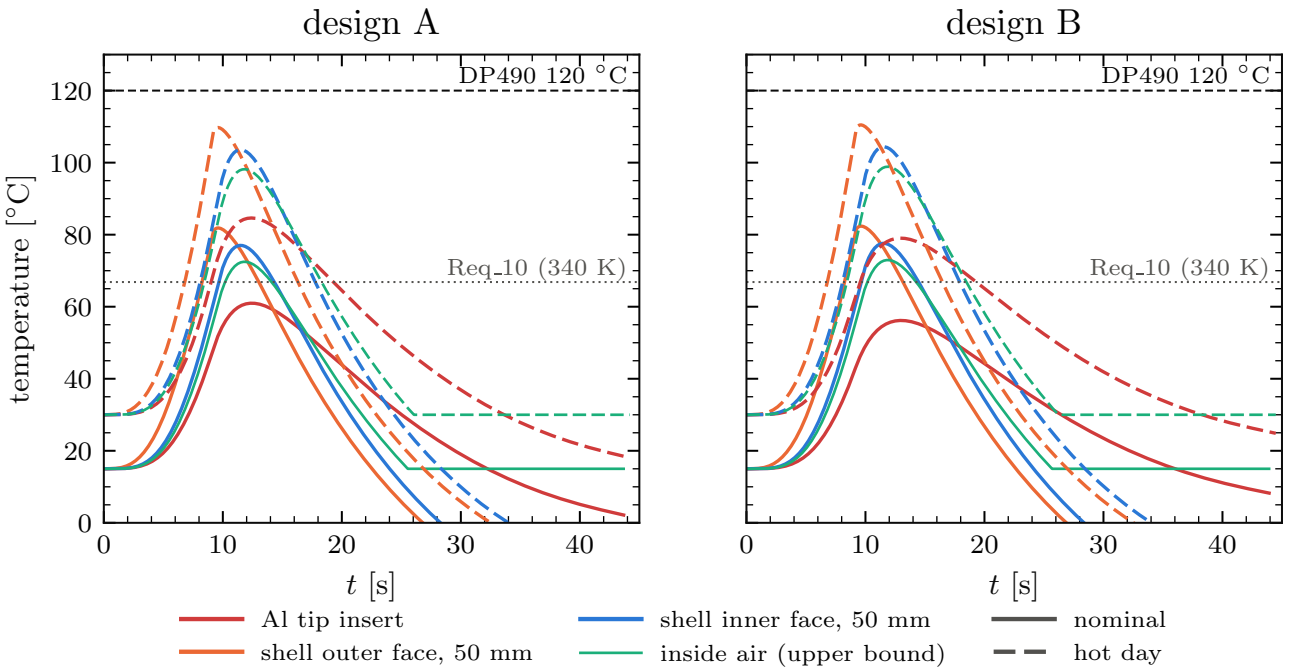


Figure 11: Temperatures during the flight, design A (a) and B (b): aluminium tip insert, CFRP shell at 50 mm (outer and inner face) and the inside air (upper bound). Solid: nominal day; dashed: hot day.

Table 5: Highest stagnation, tip-insert and inside-air temperatures during the flight (the shell is in Table 4). ✗: above the limit. Tip insert: Al 6082, 60 g over the first 80 mm.

		Design A		Design B		
	Unit	Nominal	Hot day	Nominal	Hot day	Limit
Stagnation temperature of the air	°C	118	154	119	156	
Aluminium tip insert	°C	61	85	56	79	
Inside air, upper bound	°C	72	98	73	99	67 (Req_10) ✗

4 Drag: design A against design B

In short

- **Below Mach 1 the two noses have the same drag.** About 90 % of it is skin friction, which depends on the wetted surface, and that is almost the same for both.
- **Above Mach 1 design B has 13 to 15 % less drag.** The drag is then mostly shock (wave) drag, which comes from the front of the nose, and B's front is thinner.
- At burnout Fluent gives 282.6 N for A and 237.6 N for B. In flight, with a rough wall, the drag peaks at 278 N (A) and 234 N (B), 310 and 260 N on a hot day.
- The lower drag gives B a higher apogee: 8982 m against 8887 m (nominal day).

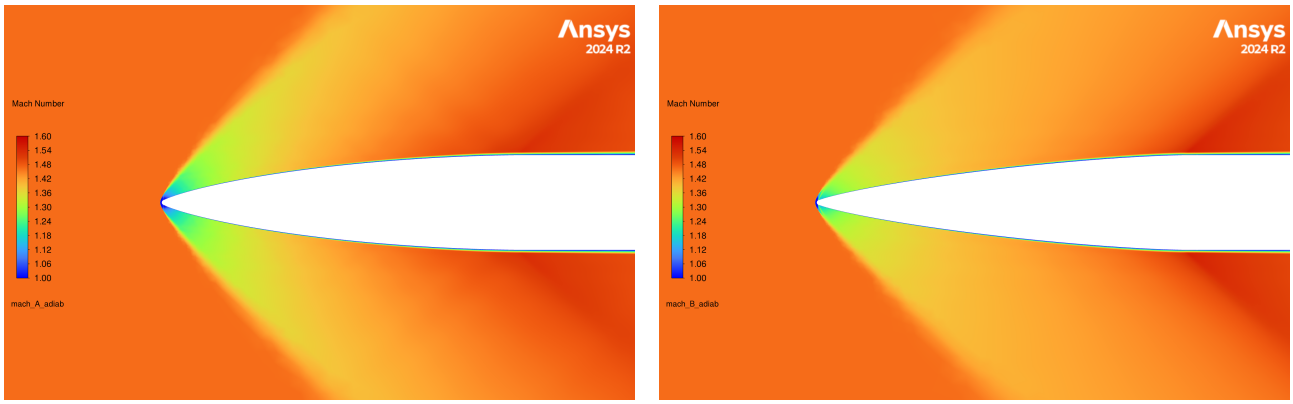
4.1 Drag at the three Fluent points

Table 6: Drag of the nose from Fluent [N], smooth wall. Pressure: the pressure drag of the curved part, mostly wave drag above Mach 1. The coupler only has friction. C_D: drag coefficient of the curved part, on $A_{ref} = \pi R^2$.

	$M = 0.95$		$M = 1.20$		Burnout, $M = 1.465$	
	A	B	A	B	A	B
Pressure drag, nose	4.6	6.0	149.2	125.2	204.7	165.2
Friction, nose	42.0	40.5	57.1	54.9	72.4	69.8
Friction, coupler	2.9	1.6	4.4	2.2	5.4	2.7
Total	**49.6**	**48.2**	**210.6**	**182.3**	**282.6**	**237.6**
C_D of the nose	0.0320	0.0320	0.0949	0.0828	0.0933	0.0791
B against A	+0 %		-13 %		-15 %	

4.2 Why B has less drag: the flow at burnout

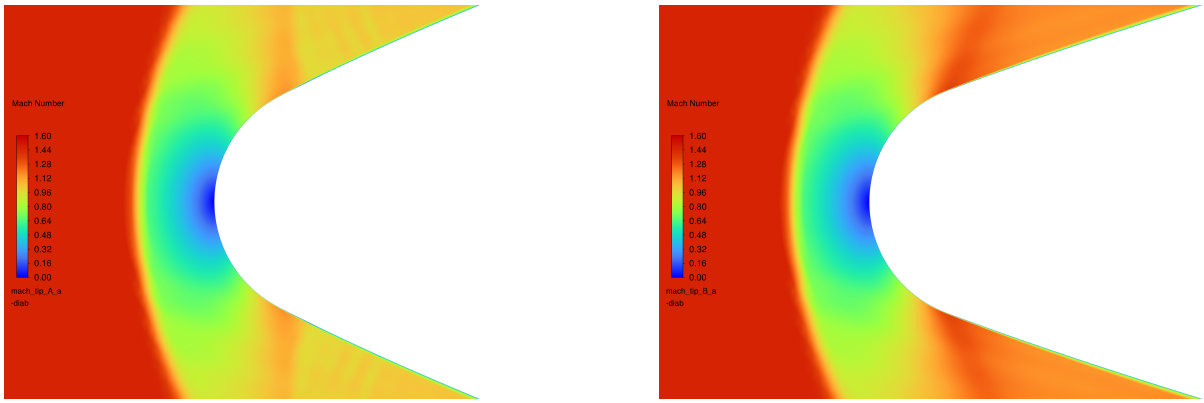
At burnout a bow shock stands in front of each tip and the air slows down around the front of the nose (Figures 12 and 13). The fuller front of design A slows the air more: it keeps a higher pressure on the front of the nose (Figure 14), and this pressure, acting on the forward-facing surface, is the wave drag. Over the first centimetres A has about twice the pressure of B (Figure 15). Further back both noses are close to the outside pressure, and slightly below it towards the shoulder, where the air speeds up onto the cylinder.



(a) Design A.

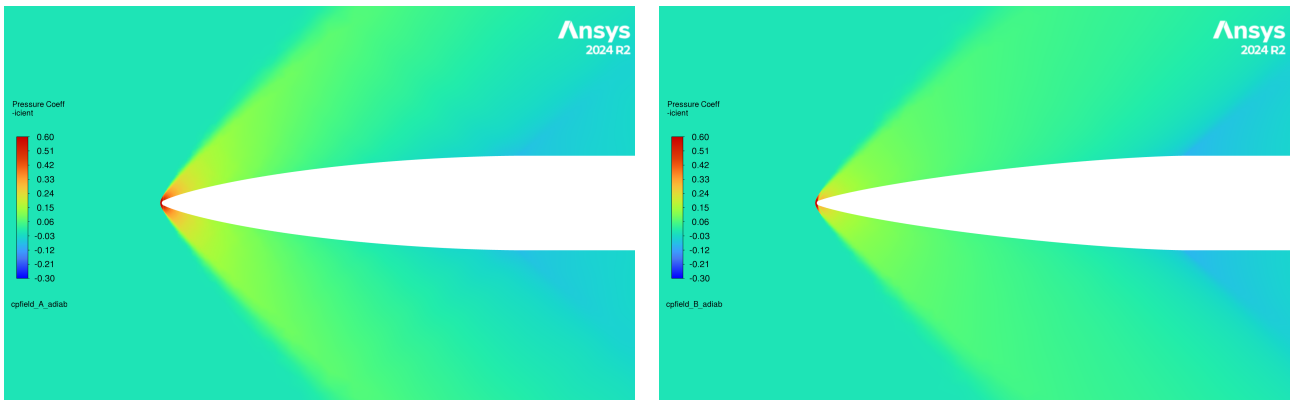
(b) Design B.

Figure 12: Mach number around the nose at burnout (Fluent). Yellow and green: the air slowed down by the bow shock and the front of the nose.



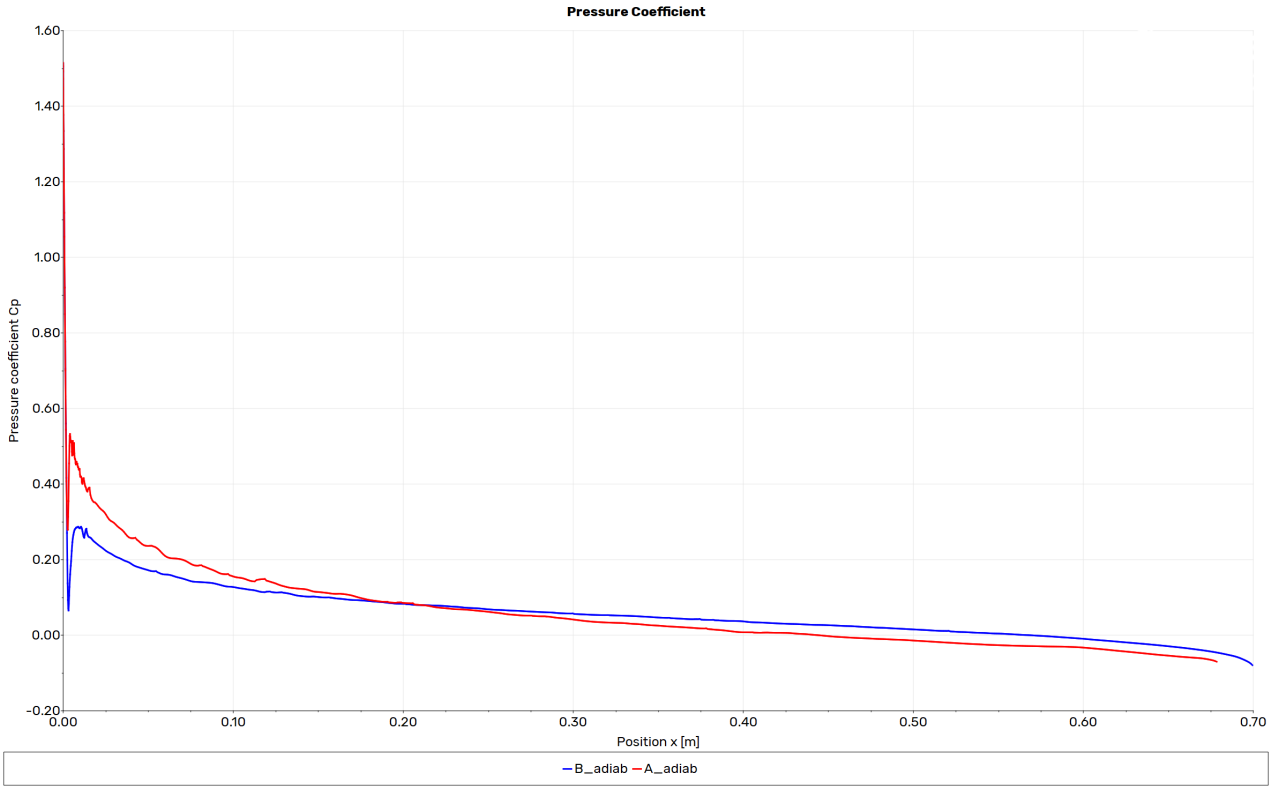
(a) Design A. (b) Design B.

Figure 13: Mach number around the tip at burnout (Fluent). The bow shock stands just ahead of the rounded tip, with a small subsonic region behind it.

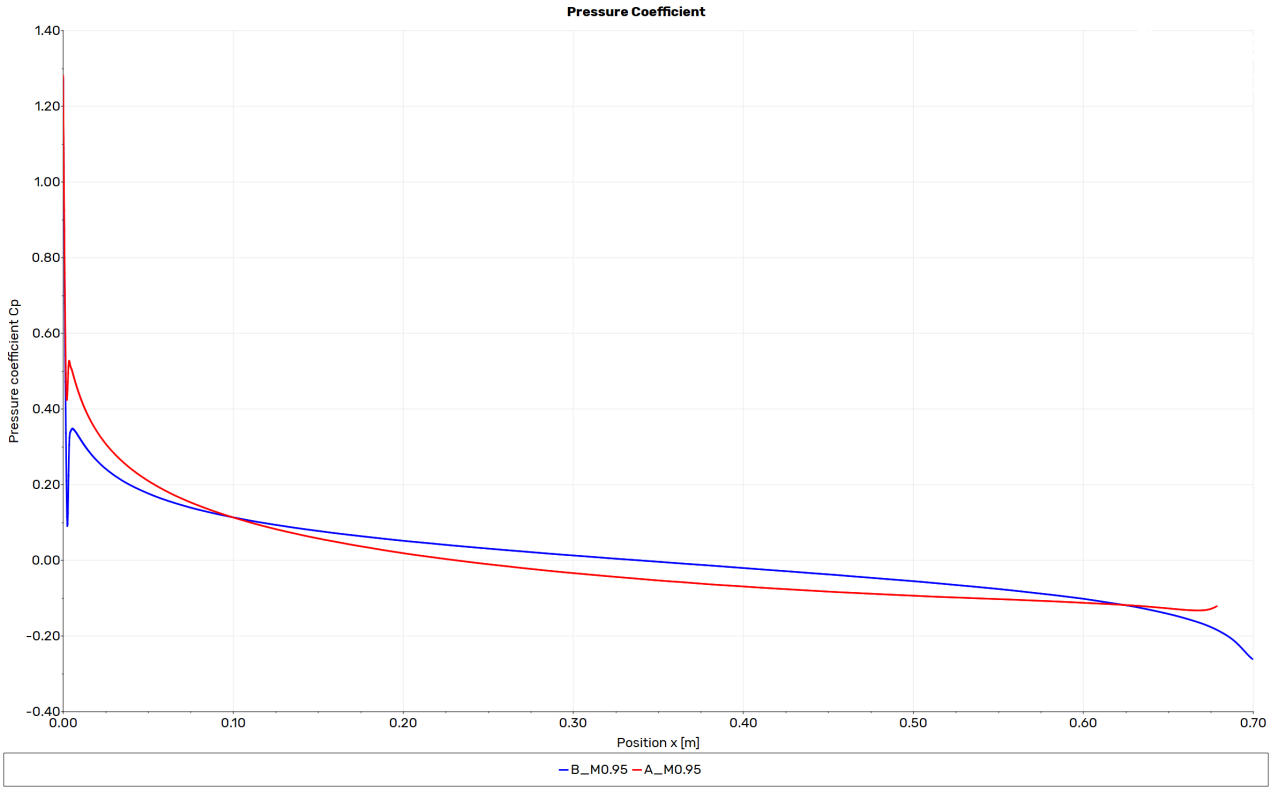


(a) Design A. (b) Design B.

Figure 14: Pressure coefficient around the nose at burnout (Fluent), $C_p = (p - p_\infty)/q$. Warm colours: pressure above the outside pressure, which pushes the nose back.



(a) Burnout, $M = 1.465$.

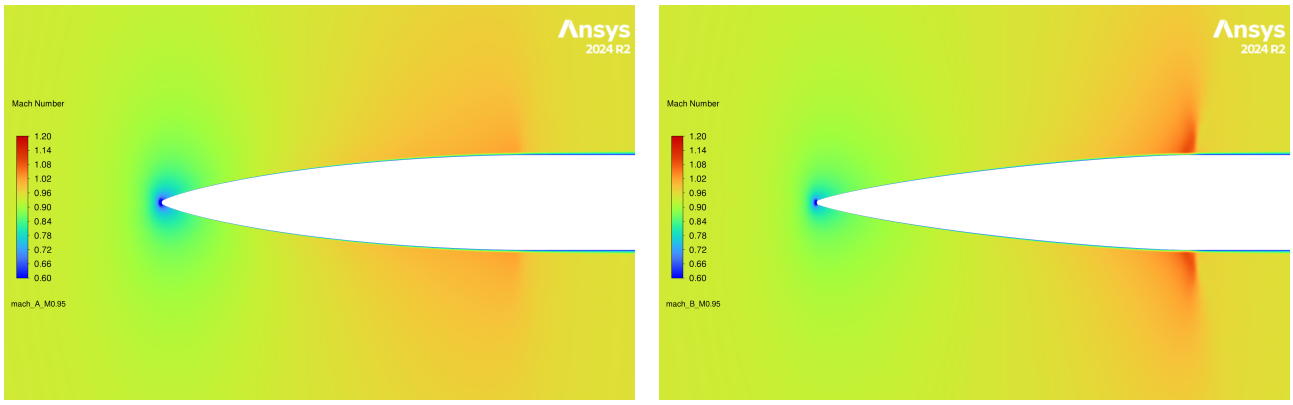


(b) Transonic, $M = 0.95$.

Figure 15: Pressure coefficient along the wall, design A (red) and B (blue), Fluent. The $M = 1.20$ run lies between the two.

4.3 Below Mach 1: same drag

At $M = 0.95$ no bow shock forms ahead of the nose (Figure 16). The pressure drag is only about 5 N for both designs (Table 6), so the drag is almost all skin friction. The friction along the wall is nearly the same for both noses (Figure 17), so their drag is the same.



(a) Design A. (b) Design B.

Figure 16: Mach number around the nose at $M = 0.95$ (Fluent). The air speeds up over the shoulder.

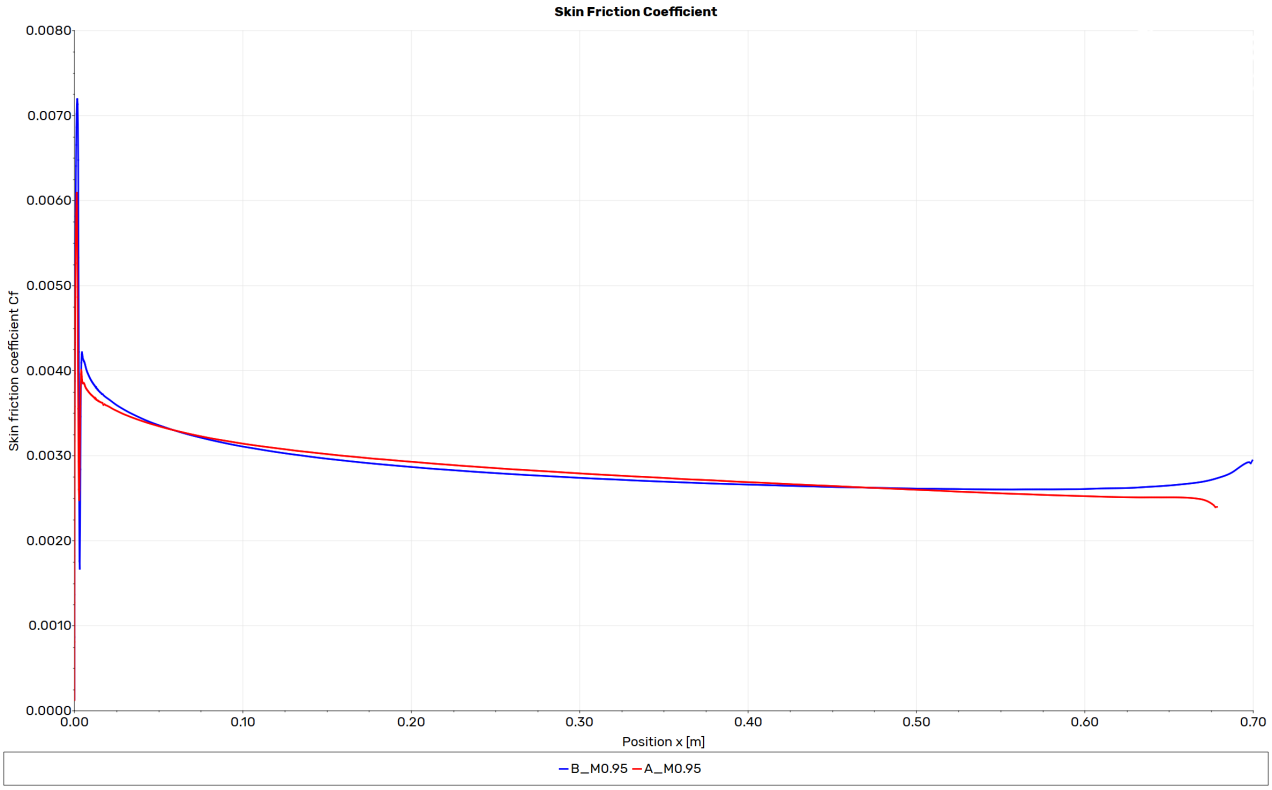


Figure 17: Skin-friction coefficient along the wall at $M = 0.95$, design A (red) and B (blue), Fluent.

4.4 Drag over the whole flight

The flight model uses a drag curve against Mach number. With the start of the drag rise moved from $M = 0.8$ to 0.94 (report 2 used 0.8), it matches Fluent at all three points within 5 % (Figure 18; details in

Appendix A.2). The drag force then follows the dynamic pressure and peaks at burnout (Figure 19). With the 20 µm roughness of the flight model the burnout drag is within 2 % of Fluent's smooth wall.

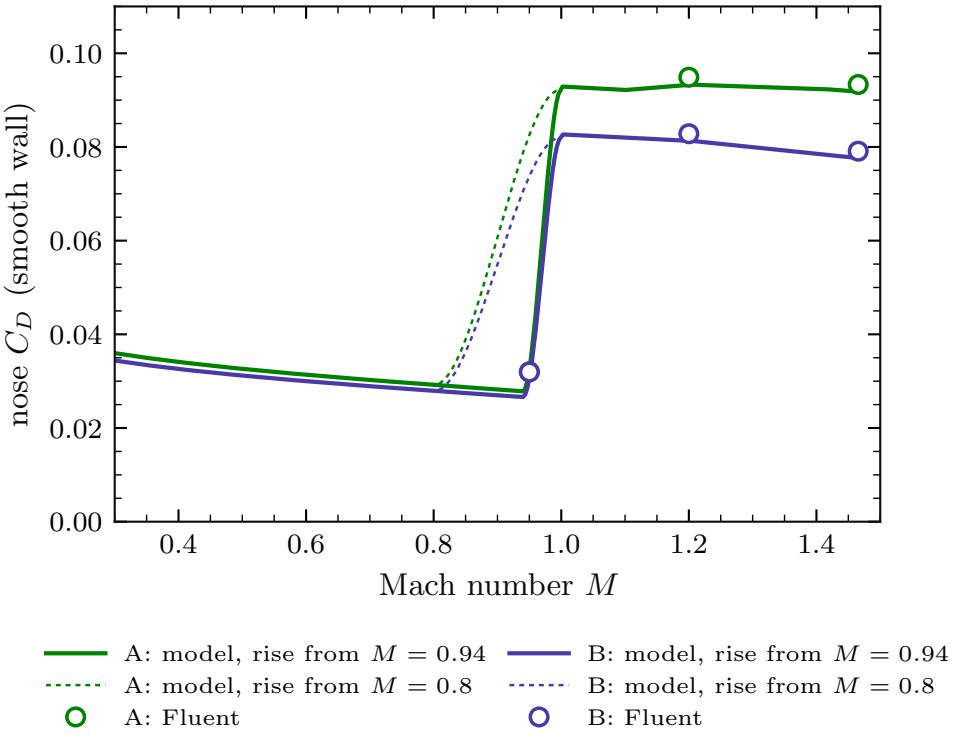


Figure 18: Drag coefficient of the nose against Mach number: flight model with the old (dotted) and new (solid) start of the drag rise, and Fluent (circles). Design A in green, B in purple; smooth wall in both.

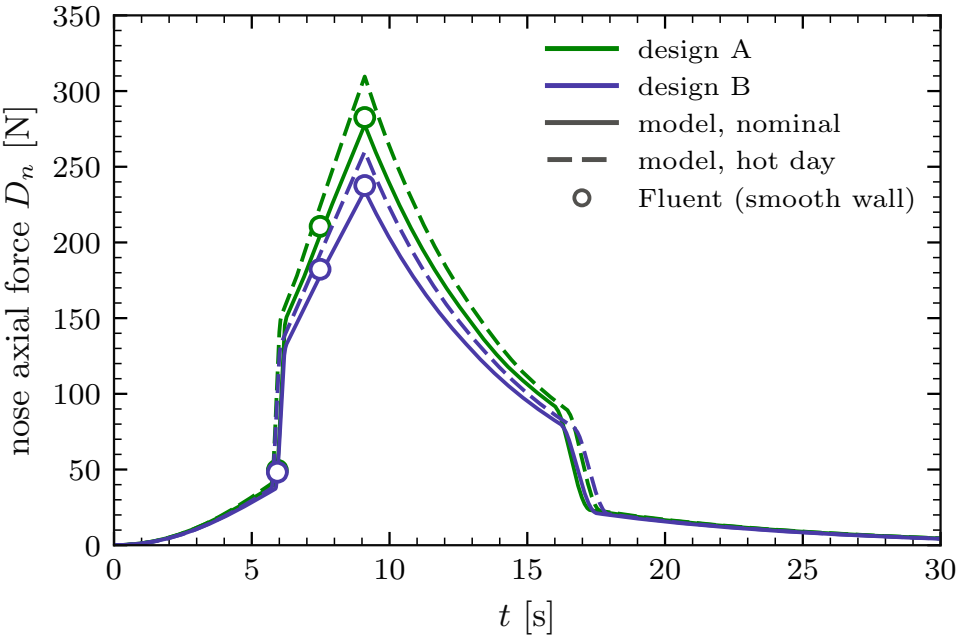


Figure 19: Drag of the nose during the flight (flight model, rough wall) and the Fluent points (smooth wall).

5 Loads on the structure

In short

- The pressure pushes the shell **inward only at the front**: on the rounded tip at burnout (up to 177 kPa), and on design A over the first few centimetres (up to 43 kPa). From mid-length to the shoulder the net pressure is **never inward**.
- The **outward** pressure grows during the whole climb as the outside air gets thinner. At apogee it is about 70 kPa on the whole shell, for both designs.
- **Along the axis, the sealed nose is pulled off the joint at every load case.** The overpressure inside outweighs the drag already at burnout, and at apogee the pull reaches about 1802 N.
- **Side force**: 104 N per degree of angle of attack at burnout (118 N on a hot day), the same for both designs. It acts 295 mm (A) or 347 mm (B) behind the tip.

5.1 Pressure along the nose, at the three load cases

At the tip the oncoming air is stopped and presses hard on the rounded cap. The pressure falls quickly over the first few millimetres. Further back the wall pressure is close to the outside pressure, and slightly below it towards the shoulder, where the air speeds up onto the cylinder. The inside stays at the pad pressure, so the rear half of the shell is pushed outward at every load case (Figure 20). Over the first 10 cm design A, with its fuller front, has about twice the pressure of design B.

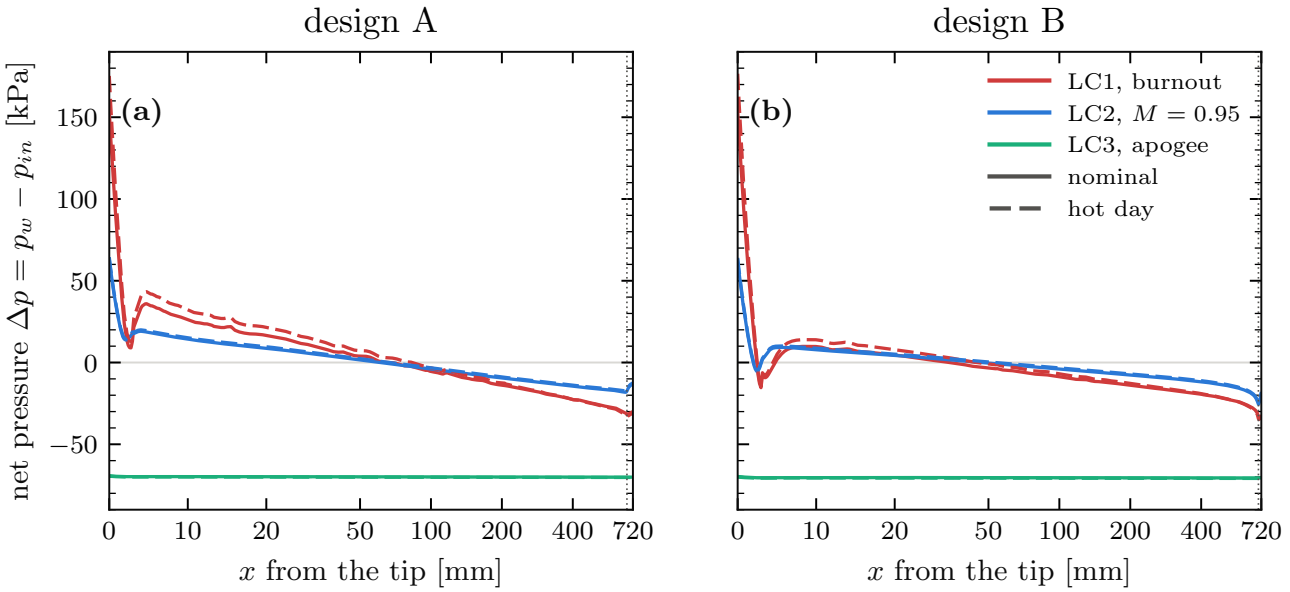


Figure 20: Net pressure along the nose at the three load cases, design A (a) and design B (b). Above zero the shell is pushed inward, below zero outward. Solid: nominal day; dashed: hot day. The x axis is stretched near the tip (logarithmic beyond 20 mm); the dotted line is the shoulder.

Table 7: Net pressure at the three load cases [kPa]. Positive: inward, with cold inside air. Negative: outward, with heated inside air. Each bound is the one that gives the larger load.

	Design A		Design B	
	Nominal	Hot day	Nominal	Hot day
LC1, burnout				
On the rounded tip (inward)	153	175	154	177
Highest behind the tip, $x > 2$ mm (inward)	36	43	17	22
Lowest on the shell (outward)	-46	-49	-48	-50
LC2, transonic				
On the rounded tip (inward)	63	64	62	64
Highest behind the tip, $x > 2$ mm (inward)	19	20	9	10
Lowest on the shell (outward)	-22	-21	-29	-28
LC3, apogee				
Whole shell (outward)	-70	-70	-71	-71

5.2 Pressure during the flight, at four stations

Figure 21 follows the net pressure at the four stations of Figure 2 through the whole flight, and Table 8 gives the extremes.

- The tip is pushed inward from lift-off until 15–17 s, with the peak at burnout.
- The other stations are pushed outward for almost the whole flight, more and more as the rocket climbs.
- When the shell heats the inside air, the inside pressure rises and adds a few kPa of outward pressure during the coast (thin lines). At the shoulder of design B this gives the largest outward value of the flight, -73 kPa at 16.6 s on the hot day.

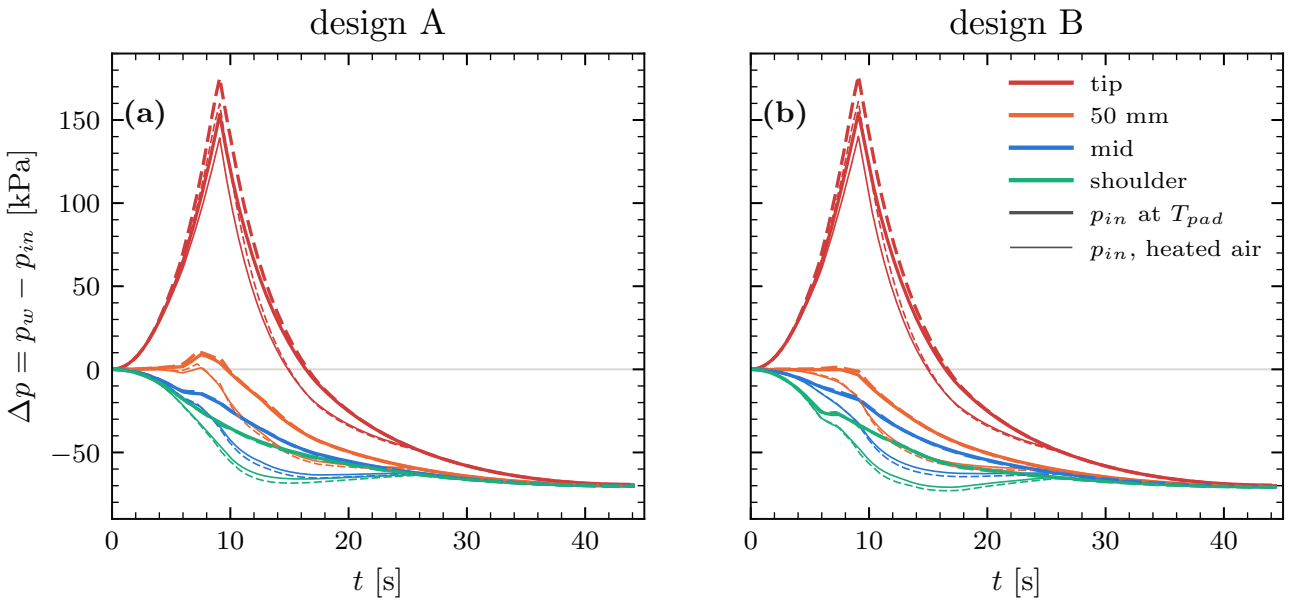


Figure 21: Net pressure during the flight at the four stations, design A (a) and design B (b). Above zero the shell is pushed inward. Thick: cold inside air; thin: heated inside air. Solid: nominal day; dashed: hot day.

Table 8: Largest net pressure over the flight at the four stations [kPa]. Inward peaks: at the tip at burnout, at 50 mm at 7.2–7.5 s ($M \approx 1.2$); 0 means never inward. Outward peaks: at apogee, except the shoulder of design B (16.6–17.0 s).

	Design A		Design B	
	Nominal	Hot day	Nominal	Hot day
Inward (cold inside air)				
Tip	153	175	154	177
50 mm	9	11	0	2
Mid-length	0	0	0	0
Shoulder	0	0	0	0
Outward (heated inside air)				
Tip	-69	-70	-70	-70
50 mm	-70	-70	-70	-71
Mid-length	-70	-70	-71	-71
Shoulder	-70	-70	-71	-73

5.3 Axial force on the joint

Table 9 gives the **net pressure force** of the sealed nose: the outside wall pressure minus the inside pressure, summed over the shell. It contains the pressure part of the drag and the pull of the inside overpressure; friction is not in it. The **drag** of the nose, which is what a vented nose would put on the joint, is in Section 4.

Table 9: Net pressure force on the joint at the three load cases [N]. Positive: compression, the nose is pushed onto the joint. Negative: tension, the nose is pulled off the joint.

	Design A		Design B	
	Nominal	Hot day	Nominal	Hot day
LC1, burnout				
Net pressure force, cold inside air	-397	-374	-436	-418
Net pressure force, heated inside air	-753	-767	-793	-813
LC2, transonic				
Net pressure force, cold inside air	-265	-248	-263	-246
Net pressure force, heated inside air	-358	-335	-355	-333
LC3, apogee				
Net pressure force	-1783	-1788	-1794	-1802

At apogee the pull of about 1.8 kN is close to the 2000 N of Req_07. Vent holes would bring the inside pressure to the outside pressure and remove it; the air load on the joint would then be the drag only. The inertial loads at the joint and the recovery load are not in this report.

5.4 Side force under an angle of attack

Fluent was run at zero angle of attack only. For a small angle of attack α, slender-body theory gives the side (normal) force of the nose and where it acts:

$$N = 2\,q\,A_{ref}\,\alpha \quad (\alpha \text{ in rad}), \qquad M_b = N\,(x_j - x_{CP}), \tag{2}$$

with M_b the bending moment at the joint plane x_j. The force is carried mostly by the front half of the nose (Figure 22); the coupler carries none. ST chooses the design angle of attack (Req_12).

Table 10: Side force per degree of angle of attack, and its point of action.

		Design A		Design B	
	Unit	Nominal	Hot day	Nominal	Hot day
Side force at LC1, burnout	N/°	104	117	104	118
Side force at LC2, transonic	N/°	51	51	51	52
Centre of pressure x_{CP}, from the tip	mm	295		347	

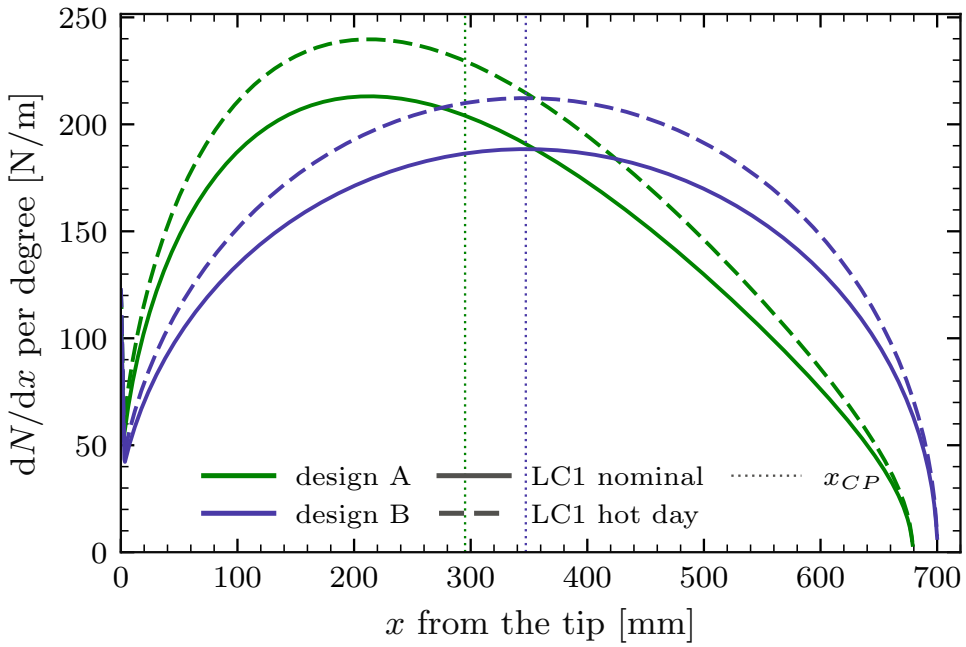


Figure 22: Side force per unit length and per degree of angle of attack at burnout (slender-body theory). Dotted: centre of pressure.

A How the numbers were obtained

A.1 Flight model and Fluent points

The trajectory comes from the Python design tool (report 2), flown with the nose of each design. Along it,

$$q = \tfrac{1}{2}\,\rho_\infty V^2 = \tfrac{1}{2}\,\gamma\, p_\infty M^2, \qquad T_0 = T_\infty \left(1 + \tfrac{\gamma-1}{2}\,M^2\right). \tag{3}$$

Nominal day: 288.15 K on the pad, propellant mass from the total impulse. Hot day: 303.15 K on the pad and the LVO masses.

The three Fluent runs use the free stream of design A's nominal trajectory (Table 11). Design B is run at the same points, so that A and B differ by shape only (B's own burnout is at $M = 1.471$).

Table 11: Free stream of the Fluent runs. Reynolds number per metre: $\rho_\infty V/\mu_\infty$.

	Transonic	Supersonic	Burnout
Time t [s]	5.93	7.48	9.10
Altitude z [m]	923	1476	2185
Mach number M	0.950	1.200	1.465
Static pressure p_∞ [Pa]	90714	84801	77685
Static temperature T_∞ [K]	282.15	278.55	273.95
Dynamic pressure q [kPa]	57.3	85.4	116.7
Reynolds number per metre [10^6/m]	20.3	24.4	27.9

Table 12: Geometry of the two noses. L: Haack length to the sharp apex; the 4.5 mm spherical tip removes δ, so the curved part is $x_S = L - \delta$ long.

	A (baseline)	B (backup)
Profile	LV-Haack ($C = 1/3$)	Von Kármán ($C = 0$)
L / δ [mm]	682.3 / 3.3	705.5 / 5.5
Curved part x_S + coupler c [mm]	679.0 + 41	700.0 + 20
Fineness L/D	3.79	3.92

A.2 Drag

The axial force is integrated over the Fluent wall faces, with r the local radius:

$$F_p = \sum (p_w - p_\infty)\, 2\pi r\, \mathrm{d}r, \qquad F_v = \sum \tau_w\, 2\pi r\, \mathrm{d}x, \qquad C_D = \frac{F_p + F_v}{q\, A_{ref}}, \quad A_{ref} = \pi R^2 = 0.025\,447\ \mathrm{m}^2. \tag{4}$$

The sums reproduce Fluent's own force report to 0.1 N. The net pressure force of the sealed nose (Table 9) is $\sum (p_w - p_{in})\, 2\pi r\, \mathrm{d}r = F_p + (p_\infty - p_{in})\, A_{ref}$.

The flight model (report 2) adds a wave drag from Euler simulations and a flat-plate friction,

$$C_{D,n}(M) = C_{D,w}(M) + C_f\,\frac{S_{wet}}{A_{ref}}, \qquad C_{D,w}(M) = C_{D,w}(1.1)\, s^2(3 - 2s), \quad s = \frac{M - M_{on}}{1 - M_{on}} \in [0,1] \tag{5}$$

below $M = 1.1$. Report 2 used $M_{on} = 0.8$, which gave 2.6 times Fluent's drag at $M = 0.95$ (0.0826 against 0.0320). With $M_{on} = 0.94$ the model matches Fluent at all three points:

Fluent / model -1	$M = 0.95$	1.20	1.465
A	-1.7 %	+1.7 %	+1.7 %
B	+4.2 %	+1.8 %	+1.9 %

The change raises the apogee by 7 m (A) and 1 m (B). Along the flight, $D_n(t) = q(t)\, A_{ref}\, C_{D,n}\big(M(t)\big)$, with the friction of a 20 µm rough wall and the coupler included.

A.3 Side force

Under a small angle of attack α [rad], slender-body theory gives, with $S(x) = \pi r(x)^2$ and V_n the volume of the nose,

$$\frac{\mathrm{d}N}{\mathrm{d}x} = 2\,q\,\alpha\,\frac{\mathrm{d}S}{\mathrm{d}x}, \qquad N = 2\,q\,A_{ref}\,\alpha, \qquad x_{CP} = x_S - \frac{V_n}{A_{ref}}. \tag{6}$$

A.4 Heating

From the two Fluent runs at burnout (adiabatic wall, wall at 300 K):

$$h(x) = \frac{q_w(x)}{T_{aw}(x) - 300\,\mathrm{K}}, \qquad r(x) = \frac{T_{aw}(x) - T_\infty}{T_0 - T_\infty}. \tag{7}$$

The recovery factor r is 0.912 (A) and 0.914 (B) on the shell.

The flight model of report 1 uses

$$h_s = 0.763\,\mathrm{Pr}^{-0.6}\sqrt{\rho_s\,\mu(T_0)\,\frac{\mathrm{d}u_e}{\mathrm{d}x}}\;c_p, \quad \frac{\mathrm{d}u_e}{\mathrm{d}x} = \frac{1}{r_n}\sqrt{\frac{2(p_0 - p_\infty)}{\rho_s}} \quad \text{(tip, Fay–Riddell)}, \tag{8}$$

$$h_t = \mathrm{St}\,\rho^* V c_p, \quad \mathrm{St} = 0.0296\,\mathrm{Re}_x^{*\,-0.2}\,\mathrm{Pr}^{-2/3}, \quad T_r = T_\infty\left(1 + r\,\tfrac{\gamma-1}{2}M^2\right),\ r = \mathrm{Pr}^{1/3} \quad \text{(shell, Eckert)}, \tag{9}$$

with ρ^*, Re_x^* at Eckert's reference temperature $T^* = T_\infty(1 + 0.032M^2 + 0.58(T_w/T_\infty - 1))$.

At burnout the model and Fluent can be compared directly (Figure 23). Without any correction the model is already within 6 % of Fluent at the tip and along the shell. Three small factors bring it onto Fluent (Table 13).

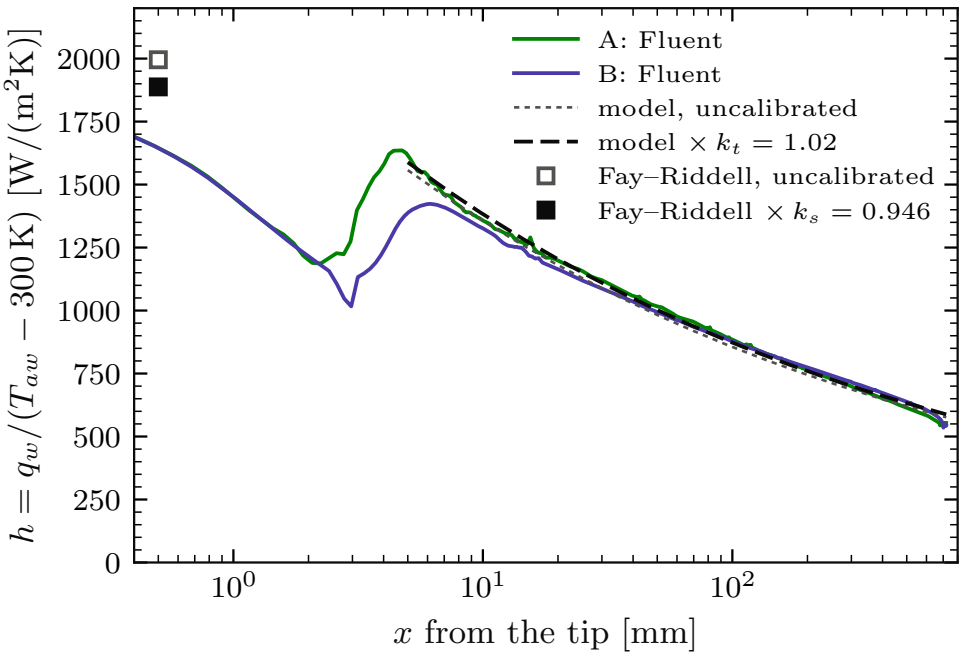


Figure 23: Heat-transfer coefficient at burnout on a 300 K wall: Fluent for A and B (smoothed between 3 and 15 mm), the model of the shell before and after calibration (lines), and the model of the tip before and after calibration (squares, placed at 0.5 mm).

Table 13: Fluent against the uncalibrated model at burnout, wall at 300 K, and the factor used in the flight model.

Fluent / model	A	B	Factor
Heat flux at the tip	0.943	0.949	$k_s = 0.946$
h along the shell, 50 mm to shoulder (median, 10–90 %)	1.010 (0.97–1.03)	1.033 (1.00–1.04)	$k_t = 1.02$
Heat into the first 80 mm (tip insert) [W]	785 / 692	629 / 550	$k_{tip} = 1.14$

The lumped tip insert takes its side heating at $x_{tip}/2$, which misses the high h just behind the spherical cap; with k_s and k_t applied it still gets 13–15 % too little heat, hence k_{tip}. Fluent's recovery factor (0.912) is a little above the model's $\mathrm{Pr}^{1/3} = 0.892$, so the model's driving temperature on the shell ($T_r = 378.9$ K) is 1–4 K below Fluent's T_{aw}.

With the calibrated h, two wall models are integrated along the trajectories:

- **shell**: one-dimensional conduction through the CFRP, heated outside, adiabatic inside,

$$\rho c \frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial y^2}, \qquad -k \frac{\partial T}{\partial y}\Big|_{out} = h\,(T_r - T_w), \qquad \frac{\partial T}{\partial y}\Big|_{in} = 0; \tag{10}$$

  run at 50 mm for the histories of Section 3 and at 22 stations from 10 mm to the shoulder for the thermal profile (`report_720/thermal_profile.py`);

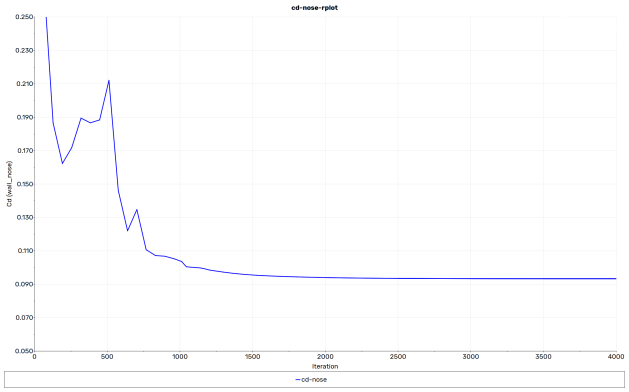
- **Al tip insert**: one temperature (aluminium conducts fast),

$$m\,c\,\frac{\mathrm{d}T}{\mathrm{d}t} = k_{tip}\left[0.55\,h_s A_{cap}(T_0 - T) + h_t(x_{tip}/2)\,A_{lat}(T_r - T)\right]. \tag{11}$$

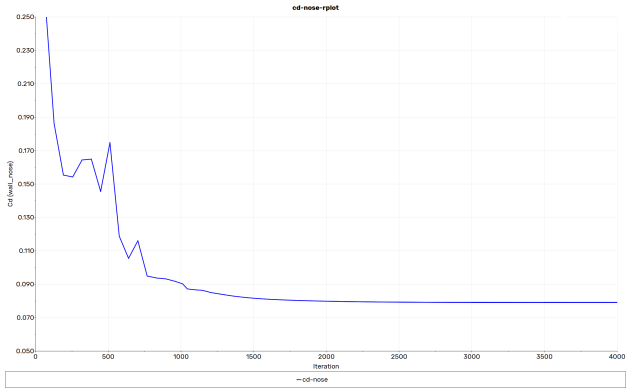
The inside air (upper bound) is the mean inner-face temperature at 50 mm, mid-length and shoulder, never below the pad temperature.

B Quality of the runs

- **Convergence**: $(\max C_D - \min C_D)/\overline{C_D}$ of `wall_nose` over the last 500 iterations is 0.007–0.050 % in all eight runs, below the 0.1 % of REPORT-4 (Figure 24).
- **Wall resolution**: over the eight runs, the largest y^+ is 0.95 on the nose and 0.39 on the coupler (Figure 26 shows burnout), so the boundary layer is resolved down to the wall as SST requires.
- **Mesh**: the REPORT-4 recipe, 174 380 (A) and 174 155 (B) cells, maximum skewness 0.90 and 0.94. No mesh study was made for these two noses; the recipe was checked on study A only.

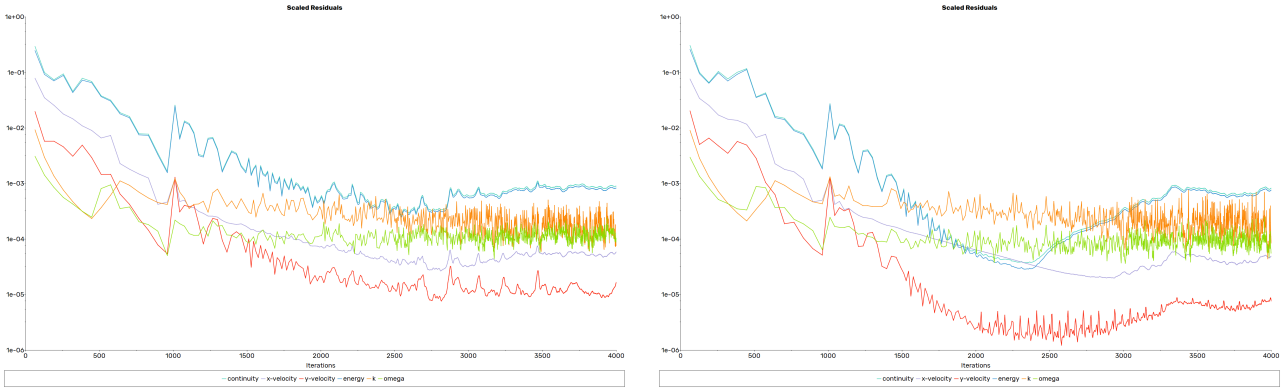


(a) Design A, burnout.



(b) Design B, burnout.

Figure 24: Drag coefficient against iterations (Fluent).



(a) Design A, burnout.

(b) Design B, burnout.

Figure 25: Residuals against iterations (Fluent).

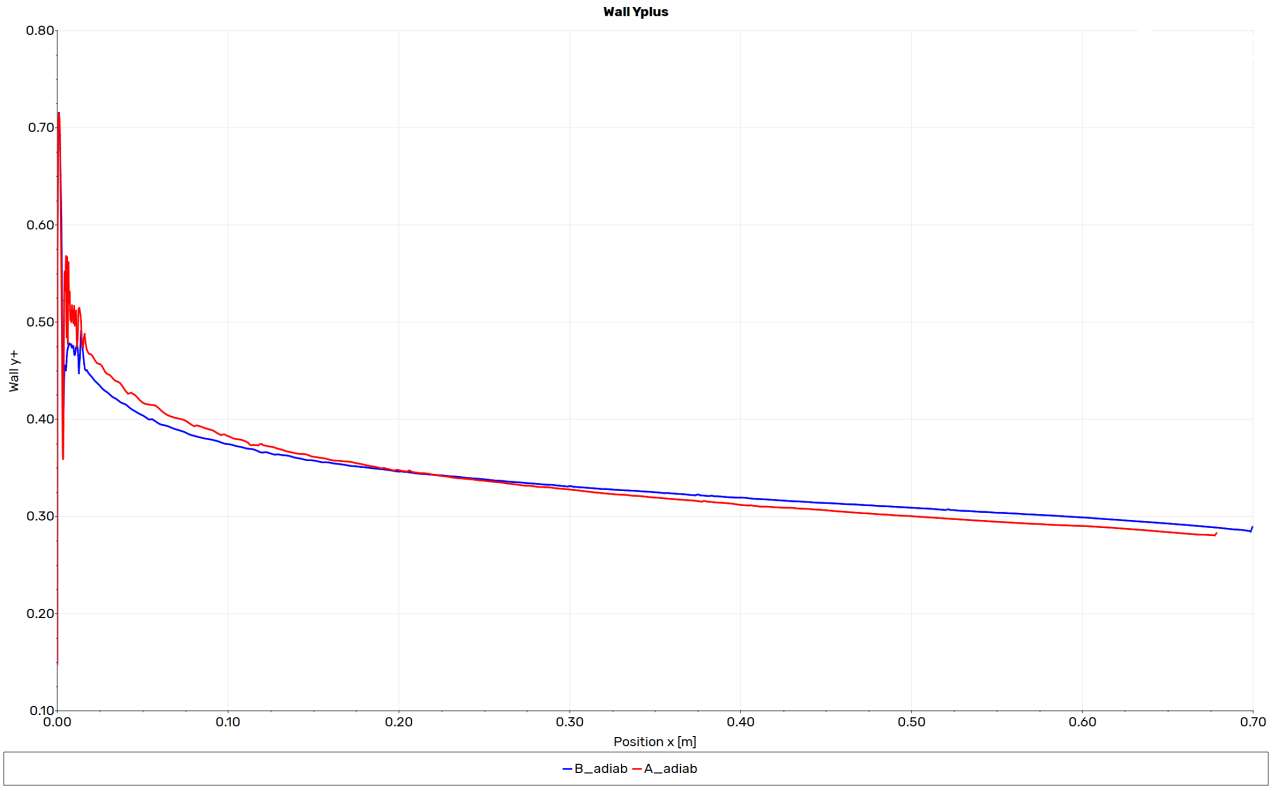


Figure 26: y^+ of the first cell along the wall at burnout, design A (red) and B (blue), Fluent.